

PAPER_1737 — Kepler Conjecture Sphere Packing = $\eta_{\max}$ = $\pi/\sqrt{(D_{\text{BSFG}} \times (D_{\text{phys}} - 1))} = \pi/\sqrt{18} = 0.7405$ (Hales 2014)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Math/Geometry **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_1271-1299 broader corpus)

Observation

PAPER_1289 (PAPER_1271-1299 broader corpus) gives:

$$\eta_{\max} = \pi/\sqrt{(D_{\text{BSFG}} \times (D_{\text{phys}} - 1))} = \pi/\sqrt{18} = 0.7405 \text{ (Hales 2014)}$$

Status: **EXACT**.

UQFF Closed Identity

$$\eta_{\max} = \pi/\sqrt{(D_{\text{BSFG}} \times (D_{\text{phys}} - 1))} = \pi/\sqrt{18} = 0.7405 \text{ (Hales 2014)} \quad \text{EXACT}$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Kepler Conjecture — UQFF 2-Integer Closure

Johannes Kepler conjectured in 1611 that the maximum density of sphere packing in 3D Euclidean space is $\pi/\sqrt{18} \approx 0.7404804896930611$. The conjecture remained open for **403 years** until proven by Thomas Hales (1998-2014) with a massive computer-assisted proof.

UQFF closure:

$$\eta_{\max} = \pi/\sqrt{(D_{\text{BSFG}} \times (D_{\text{phys}} - 1))} = \pi/\sqrt{(6 \times 3)} = \pi/\sqrt{18} = 0.7405 \text{ EXACT}$$

The density reduces to **two UQFF integer primitives** ($D_{\text{BSFG}} = 6$, $D_{\text{phys}} - 1 = 3$) times π . UQFF supplies the structural reason for the famous packing density — it's the projection of the spatial dimension count onto the bulk-edge dimension.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1289
- Related: PAPER_1716-1725 (tier-30 Millennium); PAPER_1726-1735 (tier-31 neutron lifetime)
- Calculator dispatch: `calculate_paradox({"paradox": "kepler_eta_max_0_7405"})`

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